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9/27/26

Q2.1

The pings to R4 G0/1 (10.10.10.1) and R2 G0/1(10.20.20.1) were successful because those networks are directly connected to R3. The pings to 10.1.1.4, 10.1.1.1, 172.16.40.1 and 172.16.41.1 failed because R3 does not yet have routes to those remote networks. RIP has not been configured yet so R3 only knows its directly connected networks/

Q2.2

The pings R4 G0/0 (10.2.2.4) and R2 G0/0(10.1.1.2) were successful because they are on the same directly connected 10.1.1.0/24 network as R1. The other pings failed because R1 does not yet know routes to those remote networks. No dynamic routing protocol has been configured yet

Q2.3

After RIP v2 configured, all of the pings were successful. The directly connected destinations still worked and the the previously unreachable networks became reachable because RIP allowed the routers to exchange routing information and learn routes to remote networks. This is the change from Q2.1, where only directly connected networks were reachable.

Q2.4

After RIP v2 was configured, all eight pings from R1 were successful. The remote networks that failed in Q2.2 are now reachable because RIP exchanged routing information between the routers and added routes to those networks in R1's routing table.

Q2.5

All pings from both R1 and R3 were successful after disabling auto-summary. There was no change in connectivity compared with Q2.3 and Q2.4. RIP is now advertising the more specific subnet routes instead of automatically summarizing them, but the routers still have valid routes to all of the tested networks.

Q2.6

Yes. R1, R3, and R4 all have routes with multiple next hops. R1 has equal-cost routes to 10.3.3.0/24 and 192.168.30.0/24 through R2 and R4. R3 has equal-cost routes to networks such as 10.1.1.0/24, 172.16.40.0/24, and 172.16.41.0/24 through R2 and R4. R4 has two equal-cost routes to 10.20.20.0/30, one through R3 (10.10.10.2) and one through R2 (10.1.1.2). RIP installs both because the paths have the same hop-count metric.

Q2.7

There were no significant changes in the routing tables on R1, R3, or R4 after changing the bandwidth on the R3-R4 link to 10 Mbps. The bandwidth command did not affect RIP route selection because RIP uses hop count as its routing metric, not bandwidth. Therefore, the same routes and equal-cost paths remained in the routing tables.

HOURS TO COMPLETE WAS 3 Hours